

Lecture 23: Eigenvalue and Eigenvector (second part)

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[Based on slides for „Coding the Matrix“]



WEEK

TWELVE



Agenda

Chapter 12

Lecture 23:

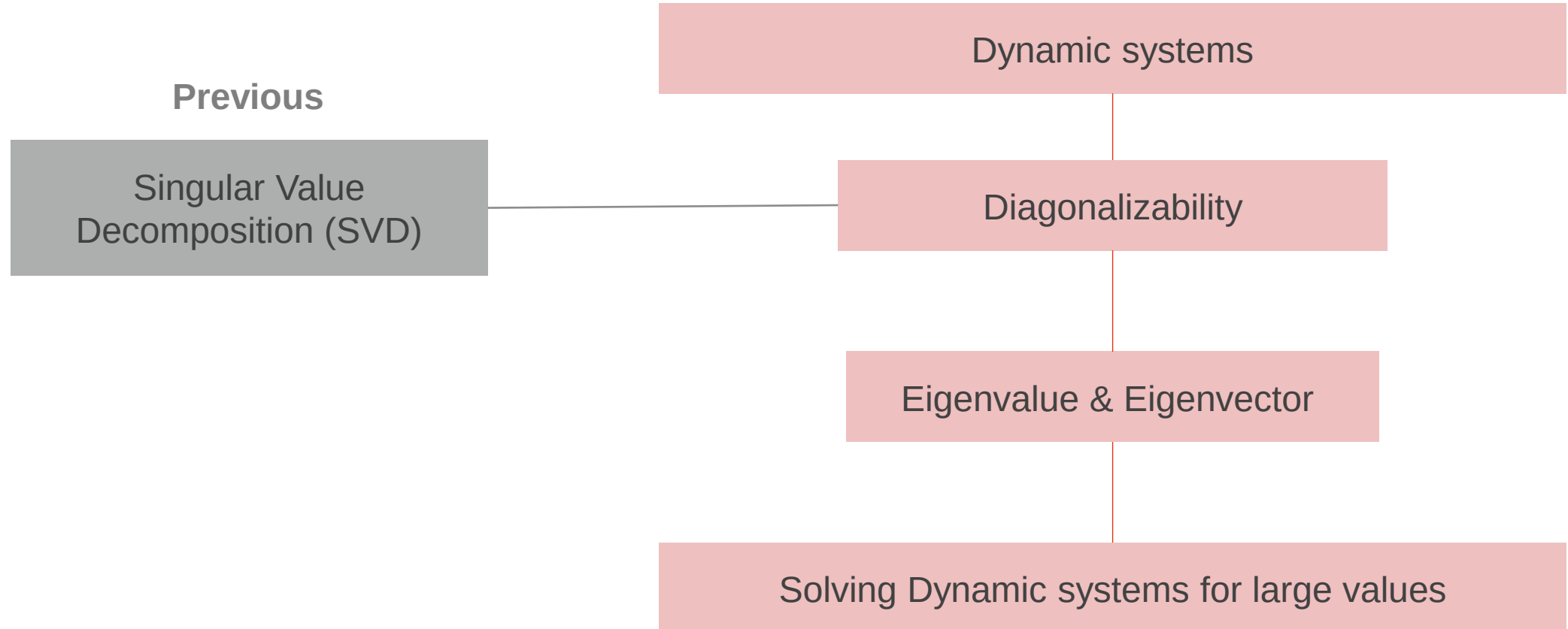
Positive-definite and positive-semidefinite matrices
Power Method
Markov chain
Stationary distribution





Lecture 22 - Summary

Concept & Definition



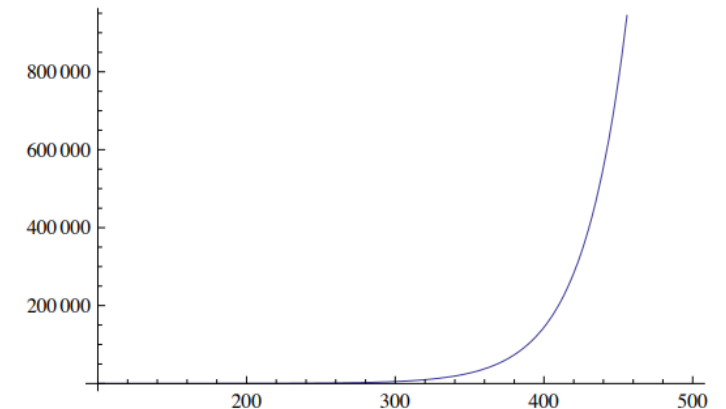


The Internet Worm

- Program that reproduces over the network (internet)
- Worms have 10% chance of reproducing in connected computer
- If a worm is already in a computer, the new worm flags itself dead in the next cycle
- With probability 1/7 the worm flags itself immortal instead

$$A = \begin{bmatrix} 0 & 0 & 1/10 & 1/10 & 1/10 & 1/10 \\ 1/7 & 1 & 0 & 0 & 0 & 0 \\ 1/10 & 1/10 & 0 & 0 & 1/10 & 1/10 \\ 0 & & 1/7 & 1 & 0 & 0 \\ 1/10 & 1/10 & 1/10 & 1/10 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/7 & 1 \end{bmatrix}$$

$$\lambda_1 = 1.034$$





Positive-definite and positive-semidefinite matrices

$$\begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} U \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{bmatrix} \begin{bmatrix} V^T \end{bmatrix}$$

$$A^T A =$$



Other nice properties

Definition: A symmetric matrix whose eigenvalues are all positive real numbers is called a positive-definite matrix.

Lemma: For a matrix A , for any set of T of distinct eigenvalues, the corresponding eigenvectors are linearly independent.

Theorem: A $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Theorem: Let A be a symmetric matrix over $\mathbb{R}$. Then there is an orthogonal matrix Q and a real-valued diagonal matrix Λ such that $Q^T A Q = \Lambda$.

Lemma: The diagonal elements of an upper-triangular matrix U are the eigenvalues of U .

Power Method



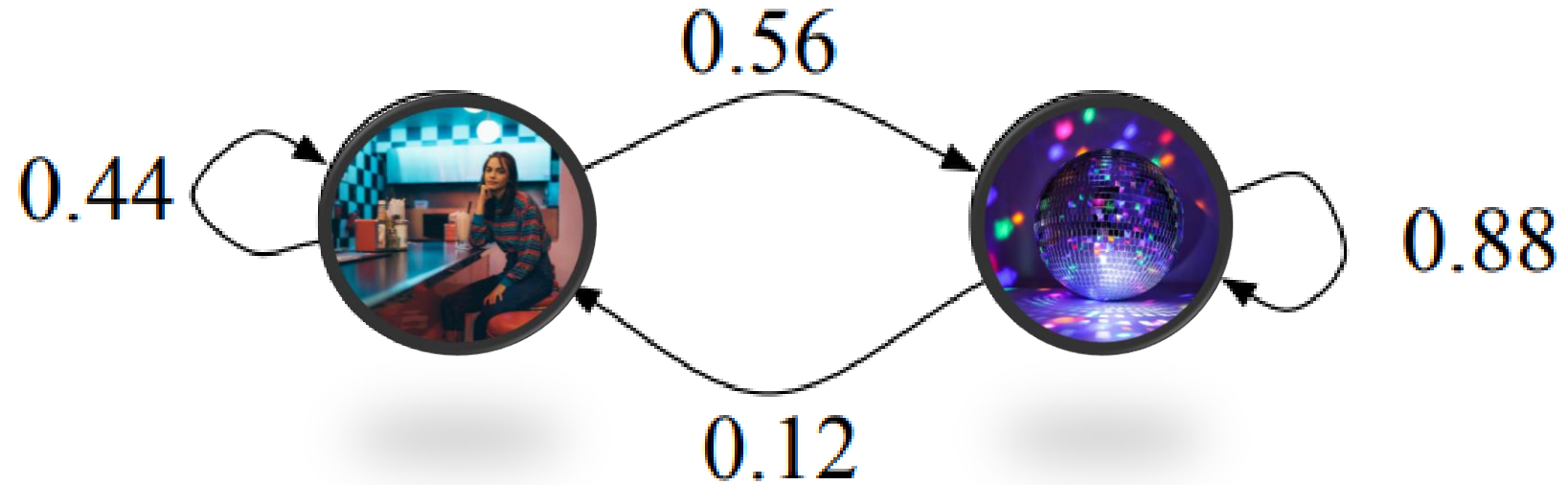
The most efficient methods for computing eigenvalues and eigenvectors are beyond scope of this class.

However, here is a simple method to get a rough estimate of the eigenvalue of largest absolute value (and a rough estimate of a corresponding eigenvector).

Assume A is diagonalizable, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ and corresponding eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$.

$$\mathbf{x}_t = \alpha_1 \lambda_1^t \mathbf{v}_1 + \dots + \alpha_n \lambda_n^t \mathbf{v}_n$$

Modelling a population



$$\begin{bmatrix} x_1^{(t+1)} \\ x_2^{(t+1)} \end{bmatrix} = \begin{bmatrix} .44 & .12 \\ .56 & .88 \end{bmatrix} \begin{bmatrix} x_1^{(t)} \\ x_2^{(t)} \end{bmatrix}$$

Modelling Randy



$$\begin{bmatrix} x_1^{(t+1)} \\ x_2^{(t+1)} \end{bmatrix} = \begin{bmatrix} .44 & .12 \\ .56 & .88 \end{bmatrix} \begin{bmatrix} x_1^{(t)} \\ x_2^{(t)} \end{bmatrix}$$

$$\begin{bmatrix} x_1^{(t)} \\ x_2^{(t)} \end{bmatrix} = (S\Lambda S^{-1})^t \begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \end{bmatrix}$$

$$= S\Lambda^t S^{-1} \begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \end{bmatrix}$$

$$= \begin{bmatrix} 0.21 & -1 \\ 0.98 & 1 \end{bmatrix} \begin{bmatrix} 1^t & 0 \\ 0 & .32^t \end{bmatrix} \begin{bmatrix} 0.84 & 0.84 \\ -0.82 & 0.18 \end{bmatrix} \begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \end{bmatrix}$$

$$= 1^t(0.84x_1^{(0)} + 0.84x_2^{(0)}) \begin{bmatrix} 0.21 \\ 0.98 \end{bmatrix} + (0.32)^t(-0.82x_1^{(0)} + 0.18x_2^{(0)}) \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= 1^t(x_1^{(0)} + x_2^{(0)}) \begin{bmatrix} 0.18 \\ 0.82 \end{bmatrix} + (0.32)^t(-0.82x_1^{(0)} + 0.18x_2^{(0)}) \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Markov chains



Definition: A n-state Markov chain is a system such:

- for each t, the current state is one of the n states of the system.
- There is a matrix A such if for a time t the system is in the state j, then for i in 1...n the probability of transitioning at the time t+1 is A[i,j].

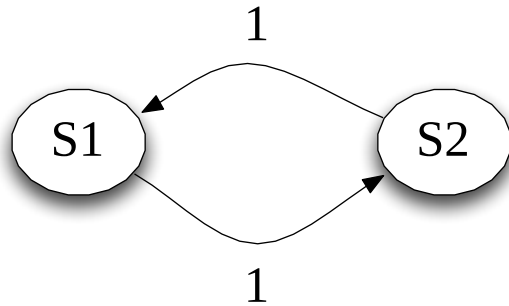
$$\begin{bmatrix} x_1^{(t+1)} \\ x_2^{(t+1)} \end{bmatrix} = \begin{bmatrix} .44 & .12 \\ .56 & .88 \end{bmatrix} \begin{bmatrix} x_1^{(t)} \\ x_2^{(t)} \end{bmatrix}$$

Stationary distributions

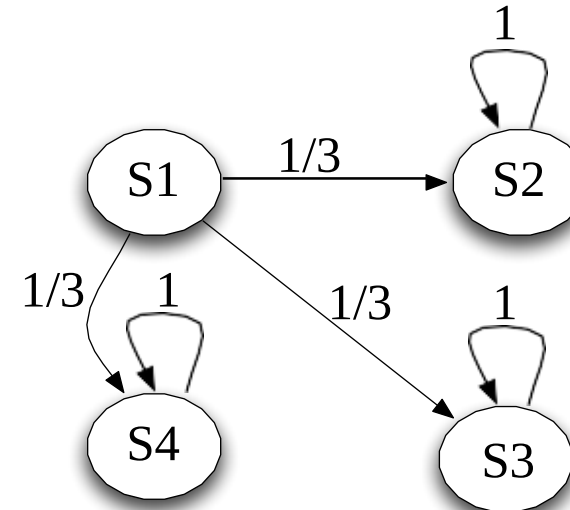


Definition: probability distribution after t iterations gets closer and closer to stationary distribution as t increases.

No stationary distributions:

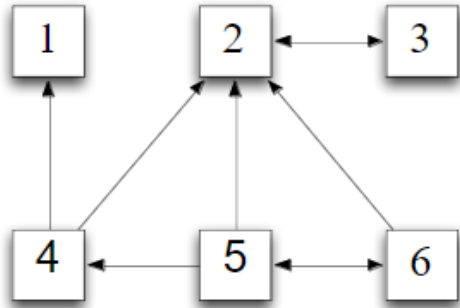


Several stationary distributions:





Modeling the web surfer: Page Rank (1)



Problem: Give the best page recommendation depending on Randy's internet behavior.

Defines a Markov chain in which the states are web pages.

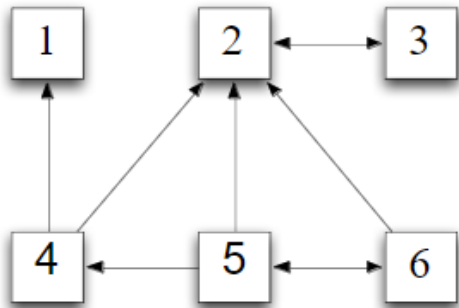
Idea: Suppose this Markov chain has a stationary distribution.

- ▶ Find the stationary distribution $\Rightarrow$ probabilities for all web pages.
- ▶ Use each web page's probability as a measure of the page's importance.
- ▶ When someone searches for “matrix book”, which page to return? Among all pages with those terms, return the one with highest probability.



Modeling the web surfer (2)

Following random links:

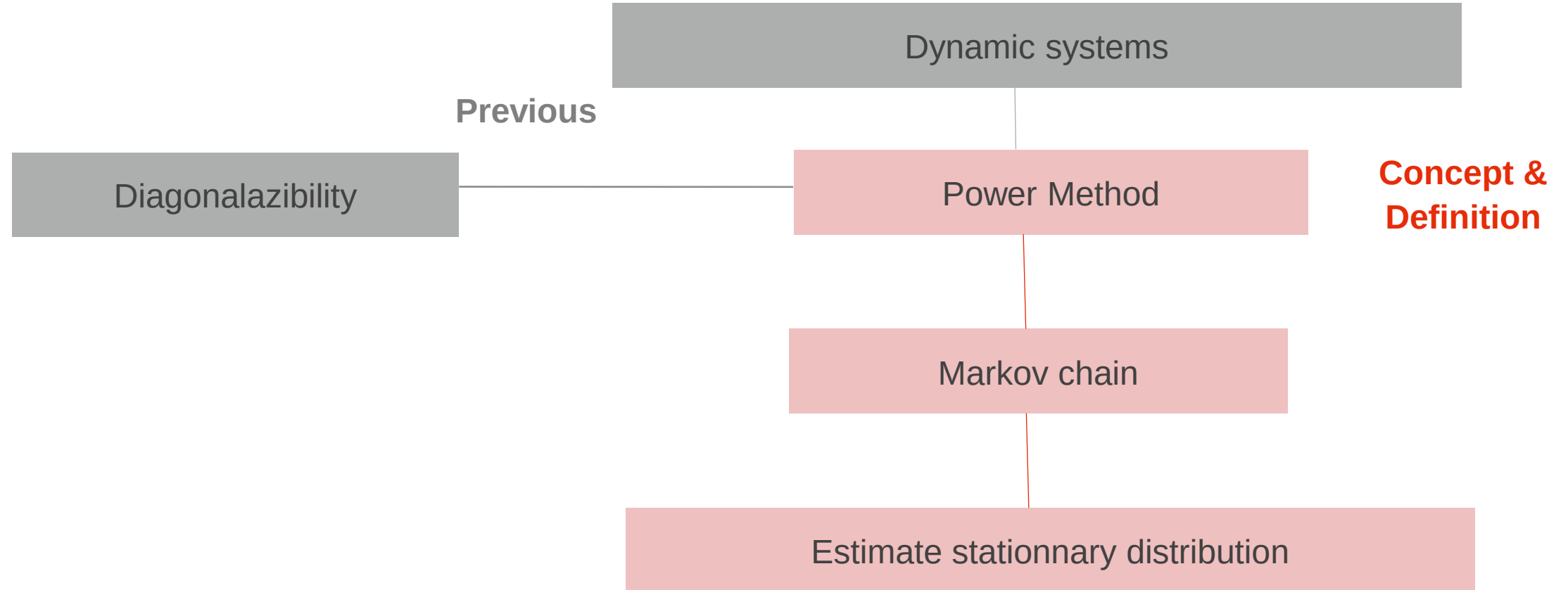
$$A_1 = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 1 & & & \frac{1}{2} & & \\ 2 & & & 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{2} \\ 3 & & 1 & & & & \\ 4 & & & & & \frac{1}{3} & \\ 5 & & & & & & \frac{1}{2} \\ 6 & & & & & \frac{1}{3} & \end{array}$$


Uniform distribution: transition matrix of the form

$$A_2 = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 2 & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 3 & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 4 & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 5 & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 6 & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \end{array}$$



Lecture **22** - Recap





Thank You For Your Attention!





Applied exercise

Problem: Can you find the eigenvalues of $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$$\lambda = \frac{1}{3} + \frac{4}{9 \sqrt[3]{\frac{\sqrt{33}}{9} + \frac{19}{27}}} + \sqrt[3]{\frac{\sqrt{33}}{9} + \frac{19}{27}} \approx 1.839286755214161$$



Applied exercise

Problem: Find the eigenvalues of $\begin{bmatrix} 2 & 1 & -4 \\ 0 & 3 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$